

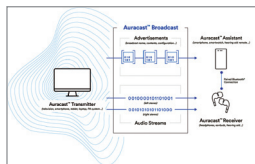
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Low-cost Bluetooth based audio broadcasting tech

Auracast is a Bluetooth based audio broadcasting technology. It enables a Bluetooth transmitter to broadcast to an unlimited number of nearby Bluetooth



audio receivers, including speakers, earphones, and other hearing devices.

Auracast can also enhance the assistive listening technology to enable hard of hearing individuals to hear better in venues where hearing aids and cochlear implants do not function well. It is a low-cost technology that is easier to deploy while offering higher audio quality and greater privacy. The technology can also enable the use of silent television in public venues like airports, gymnasiums, restaurants, etc, where it broadcasts audio that any visitor with Auracast-enabled Bluetooth earbuds or hearing aids will be able to hear.

Bluetooth

<https://www.bluetooth.com>

Defects-documenting IoT based soldering station

Ersa i-CON Trace is a first-of-its-kind soldering station with IoT capability. The soldering station can record the movement and solders a user makes.



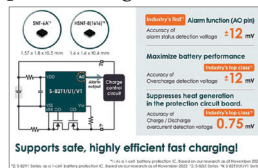
The device processes the information to identify the causes of quality deviations at an early stage, isolate them and trace problems. This can enable users

to avoid cold solder and other soldering defects and prevent device failure at a later stage. The i-CON Trace offers Wi-Fi and Bluetooth connectivity and enables users to check various parameters, including temperature, using devices such as a notebook, tablet, or smartphone. This IoT based manual soldering station can enhance manual soldering and enable users to document any defect in the solder and improve the reliability of soldered joints.

Kurtz Ers Group
<https://www.kurtzergroup.com>

Battery protection ICs with inbuilt alarm

ABLIC's S-82L1/T1/U1/V1 series, the industry's first 1-cell battery protection ICs with alarm, can accurately detect when the battery voltage reaches the preset voltage to enable safe and simple



fast charging without an AD converter. The charger ICs have one of the highest

overcharge detection voltage accuracy of $\pm 12\text{mV}$ and overcurrent detection voltage accuracy of $\pm 0.75\text{mV}$. The high accuracy of overcharging and overcurrent detection enables the ICs to detect the battery voltage even during fast charging, ensuring safe and efficient fast charging and significantly improve the product safety. The ICs are suitable for use in smartphones, tablets, smartwatches, etc.

ABLIC

<https://www.ablic.com>

Wi-Fi 7-ready platform for high-end smartphones

The Dimensity 9200 is the first Wi-Fi 7-ready smartphone platform, sup-

porting up to 6.5Gbps data rates. It is capable of providing the first 64-bit exclusive Armv9 Performance Cores for smartphones and features the fastest smartphone memory, LPDDR5X 8533Mbps. This MediaTek chipset is



the first smartphone chip to integrate an Arm Cortex X3 with operating speeds of over 3GHz. The 5G system on chip (SoC) offers enhanced

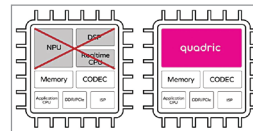
performance and better power saving than the previous generations of Dimensity chipsets. The Dimensity 9200 is also suitable for use in high-end smartphones with foldable displays. It offers extreme performance and intelligent power efficiency. It can provide an ultra-sharp image capturing and support for both mmWave 5G and sub-6GHz connectivity.

MediaTek

<https://www.mediatek.com>

GNPNPU that simplifies SoC hardware design

Chimera is the first family of general-purpose neural processors (GNPNUs) from Quadric. The GNPNPU blends the machine learning (ML) performance characteristics of a neural processing accelerator with the full C++ programmability of a modern digital signal



processor (DSP). Chimera GNPNPU provides one unified

architecture for ML inference plus pre- and post-processing, greatly simplifying system-on-chip (SoC) hardware design. The new architecture offers significant advantage as the neural network graphs and C++ code are merged into a single software code stream. Only one tool chain is required for scalar, vector, and